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NOVEMBER 16, 1903.



The Great Thames Barrage.

By

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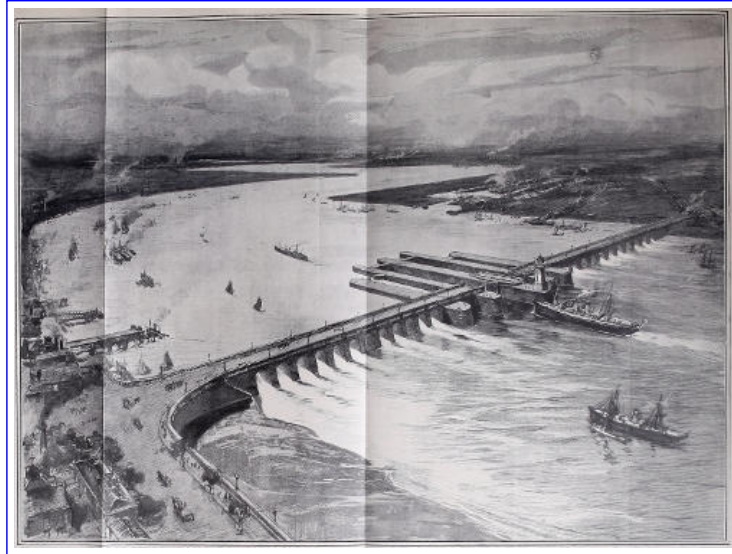
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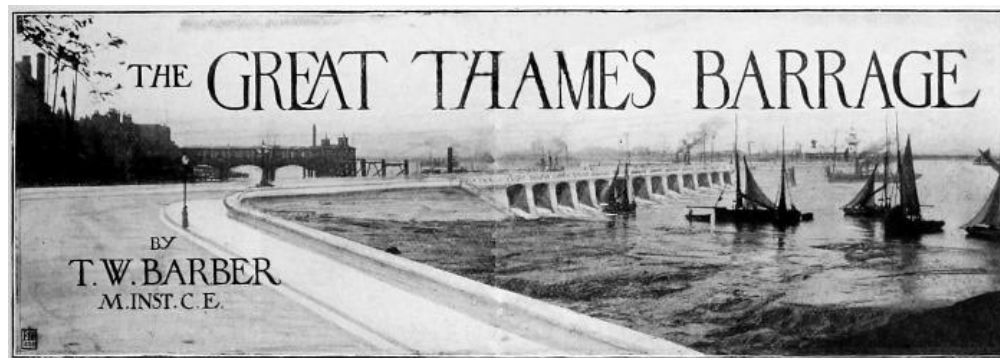
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THE PROPOSED THAMES BARRAGE: A VIEW OF THE RIVER FROM THE GRAVESEND BANK AS IT WOULD APPEAR IF THE DAM WERE CONSTRUCTED

DRAWN BY H. C. BREWER FROM MATERIALS SUPPLIED BY MR. T. W. BARBER

Mr. T. W. Barber, M.Inst.C.E., and Mr. Jas. Casey, M.I.N.A., have suggested that the difficulties of which the shipping interests complain might be met by the construction of a barrage across the river from Gravesend to Tilbury, a comparatively simple engineering feat after the great Nile dam (about $1\frac{1}{4}$ miles in length), especially as the bed of the stream is here firm chalk. This would, it is claimed, give a navigable depth of water, varying from 65ft. at Gravesend to 32ft. at London Bridge, without dredging, or any interference with the river bottom or banks. Some of the advantages which would, the advocates of the scheme claim, be secured are as follows:—Ships drawing 30ft. could proceed to London Bridge at any hour of the day or night, without waiting for tides; ships of all tonnages and draughts could traverse the river, anchor anywhere, lie alongside any wharf or quay, always remain at one level for loading or unloading, and need not lie out in the river or obstruct the free navigation; dock entrances could be left open, thus saving the cost and time lost in working them—the London and India Docks Company estimates the cost of working their entrances at £50,000 per annum; while greatly increased safety of navigation would result, there being no possibility of grounding, swinging with the tides, or collisions due to tidal drift. In addition to these, London would be provided free with a lake of fresh water forty-five miles long, and from a quarter to a half-mile wide. In short, we should have a vast inland lake from Gravesend to Richmond.



THE GREAT THAMES BARRAGE

BY
T. W. BARBER
M.INST.C.E.



It is not necessary to emphasise in any way the fact that something must be done in the tidal Thames to bring the Port of London up to date, and to maintain it as the great inlet of British commerce. What with numerous newspaper articles, magazine reviews, reports of Royal Commissions and others, and a general murmur of complaint from all persons who use the port for their business or the river for traffic purposes, there have recently been abundant evidences that things are not as they should be. Everyone is agreed on this point, but when it comes to the question of a remedy, there agreement ends and confusion begins.

And, first, to briefly catalogue the ***What is complained of.*** complaints from all sources. They are as follows:—(a) Insufficient depth of water in the river for the increasing size and tonnage of steamships. (b) Tide-waiting at Gravesend and at the dock entrances, inward and outward. (c) Excessive dues. (d) Vexatious restrictions owing to conflicting and overlapping authorities in the river. (e) Excessive cost of barging, pilotage, and labour in loading and discharging. (f) Loss of time at the port. (g) Dangerous navigation, due to tides, bends in the river, narrow channel, fogs, and the crowded state of the river. That these complaints are well founded is generally admitted.

Remedies Proposed.

The Royal Commission on the Port of London, the Board of Trade, as representing the Government, the Thames Conservancy, the dock companies and others recommend the deepening of the river by dredging as a remedy for (a), and as a partial remedy for (b) and (f). As to (c) no remedy seems to be proposed by either, but rather an increase of dues, or in lieu thereof a charge upon the rates of London through the London County Council.

Partly to amend (d) it is proposed by all the above authorities, except the Thames Conservancy, that a Port Trust should be created to control the river, instead of the present conflicting authorities of the Thames Conservancy, Trinity House, the City Corporation and the Watermen's Company.

But as to (e) there is no suggestion of amendment, nor is it expected that the proposed deepening of the river will materially improve the dangerous navigation (g).

***Port of London Bill,
1903.***

The Government has sought to give effect to the Report of the Royal Commission on the Port of London in this Bill, which reached the stage of Committee of the whole House, and was then suspended till next Session (1904).

But as there were seventy petitions presented against the Bill, and a large number of amendments stand on the notices for Committee of the whole House, it may justly be concluded that the Bill satisfies no one, and that the attempt of the Government to force it through the House by stifling discussion of most of its vital points in Committee was a flagrant violation of public rights, and will have a disastrous effect on the future settlement of the question.

***Dockisation the True
Remedy.***

In 1755 Smeaton proposed the dockisation of the River Clyde as a means of providing a sufficient depth of water for the increasing trade of the Port of Glasgow. His plan was rejected, and the Clyde Trustees have since expended £7,430,000 in dredging and improving the river to a low-water depth of 20 ft., and now spend annually a large sum in maintaining this depth.

Thos. Howard proposed the dockisation of the Avon at Avonmouth in 1877 to provide a sufficient depth of water for vessels passing to the Bristol Docks up and down the Avon, there being a rise and fall of tide in the Severn of nearly 40 ft. His proposal was not adopted because the extraordinary range of tide would have left the entrance unapproachable at low water, causing delay in the Severn Channel.

Messrs. L. Murray and W. C. Mylne recommended the dockisation of the River Wear in 1846, but this was not carried out.

The Czar of Russia has recently approved a great dockisation project, consisting of a dam with locks and sluices across the Straits of Kertch, in the Black Sea, to raise the level of the Sea of Azov for the purpose of facilitating navigation to the port of Taganrog and the River Don. The Sea of Azov will then become a fresh-water lake, with an increased depth of water (14½ ft.) and an area of 10,000 square miles. The dam will be nine miles long, and is estimated to cost £5,000,000.

There is, however, no actual instance of the dockisation of a tidal river from which any data can be obtained.

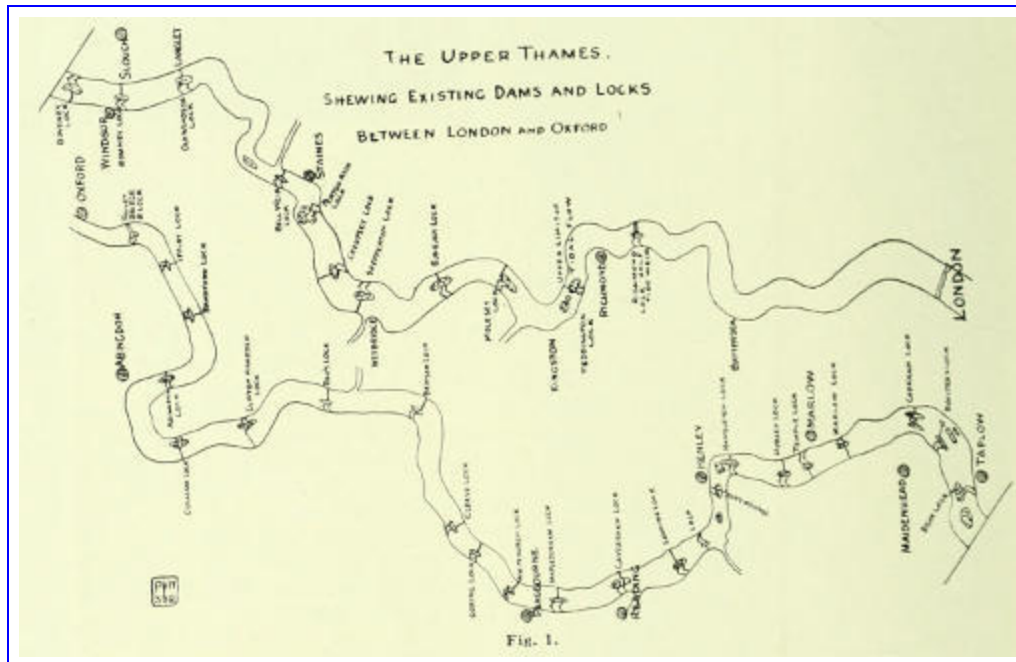


Fig. 1. THE UPPER THAMES.

SHEWING EXISTING DAMS AND LOCKS BETWEEN LONDON AND OXFORD

The Thames, moreover, differs entirely from any of the foregoing rivers, and must be considered on its own merits. The map ([Fig. 1](#)) shows that it is already dammed and provided with locks at thirty-four places between London and Oxford, the object of these dams being the maintenance of a uniform level of water for navigation and boating purposes, and to prevent the river running dry in the dry season and exposing the muddy foreshores.

But from Teddington Weir to its estuary the Thames is tidal, and there is no obstruction to the tidal flow except the bridges and the half-tide weir at Richmond, which merely holds up sufficient water to cover the foreshores for the advantage of the riparian owners and of boating.

The Tidal Thames.

To understand clearly the conditions to be dealt with, it is necessary to consider the daily movements of tide, the affluents, the dock and wharf business and the traffic of the river.

The maps ([Figs. 2 and 3](#)) show the tidal river and estuary from Teddington to the North Foreland. [Transcriber's Note: It seems 'Teddington' here is an error for either 'London' or 'Southwark'; that's what the maps show, anyway.]

The river proper—that is, from Teddington to Gravesend—is forty-six miles long, and averages one-third of a mile wide. Its depth at low water varies from 6 ft. at Teddington to 10 ft. at London Bridge and 40 ft. at Gravesend, and the rise of tide at London varies from 17 ft. to 21 ft. and at Gravesend from 15 ft. to 19 ft., the current usually averaging four knots per hour. At London Bridge the Spring tides flow 5 hours and ebb 7½ hours; while at Gravesend they flow 6 hours and ebb 6½ hours.

The river winds about considerably. The straight line distance from Teddington to Gravesend being thirty-three miles, shows that thirteen miles are added to the river in its bends, some of which—as those at Grays, Erith, Blackwall and Limehouse—are short and tortuous.

The longitudinal section ([Fig. 4](#)) of the river from Teddington to Gravesend gives graphically all the data necessary for our purpose. Ordnance Datum (O.D.) is the common datum line of the Government maps. Trinity High Water (T.H.W.) is the water datum usually adopted in the river. High and low water, ordinary and Spring tides (H.W.O.T.—L.W.O.T.—H.W.S.T.—L.W.S.T.) are the levels of the respective states of tide in the river at various points. The highest and lowest known tides are also given, as well as the level of the river bottom and the levels of the principal dock entrance sills and of the crowns of the Thames tunnels, showing their depths below the river bottom.

Tidal Wave.

The curved lines (in various forms of dotting) represent the levels of the surface of water at various states of Spring tides and clearly show the tidal wave which ascends the river and by its momentum and volume raises the high-water level at the upper end several feet above that at Gravesend.

The Thames Estuary. From Gravesend to the Nore is an immense triangular area with sandy bottom, muddy foreshores and several deep channels running in the general direction of the Essex coast line, that is, N.E. to the North Sea. The area may be roughly estimated at 120 square miles, and the navigable depth of the principal channels at from 60 ft. to 26 ft. at low water Spring tides.

The volume of the estuary at high water Spring tides may be taken at 2600 million cubic yards, and at low water Spring tides at 1500 million cubic yards, the volumes of the river from Gravesend to Teddington being respectively 180 million and 80 million cubic yards, so that the volume of tidal water entering the river each tide is about 100 million cubic yards.

Upland Water.

But there is a daily flow over Teddington weir—excluding the water abstracted by the London water companies—varying during the year on the average as follows:—

	Cubic yards.
Jan.	11,800,000
Feb.	5,300,000
March	4,100,000
April	3,250,000
May	4,720,000
June	2,900,000
July	1,760,000
Aug.	1,590,000
Sept.	1,160,000
Oct.	1,900,000
Nov.	3,530,000
Dec.	8,230,000

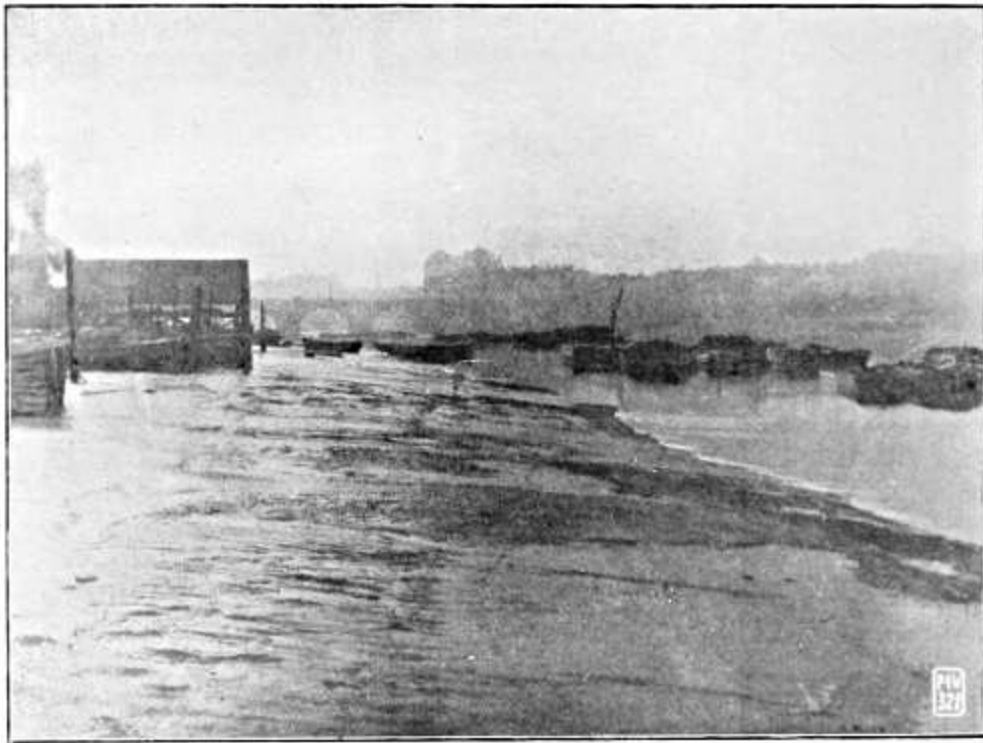
Average daily flow, 4,186,000 cubic yards.

Below Teddington, numerous small affluents add to this volume of upland water as follows:—

	Cubic yards per day.
The River Lea and Essex streams on the north bank	60,000
Streams in the Kent district	500,000
To this must be added a large quantity of spring water rising in the bed of the river and land drainage— quantity uncertain	1,000,000
Sewage effluents discharged at Crossness and Barking	1,176,000
Storm water overflow from London sewers	580,000
Total upland fresh water daily average	7,502,000

This gives an average volume of $7\frac{1}{2}$ million cubic yards of fresh water descending and mingling with the oscillating tidal water of the river and estuary, which slowly pushes the latter down into the North Sea. Taking the high-water volume in the river as above at 180 million cubic yards, the proportion of fresh water from the upland daily flow is $\frac{1}{24}$ th, and therefore it will take 24 days to change entirely the water in the tidal river.

Mr. W. P. Birch has shown that the combination of fresh water and sewage which enters the river below Teddington remains in the river, oscillating up and down with the tides for 45 days before it finally gets pushed out into the North Sea.



THAMES MUD.

In this way the discharge of effluents at Crossness and Barking passes up and down in front of London for more than a month, and it becomes apparent that the tidal action keeps the river continually saturated with about 45 days' soilage. It is no wonder, therefore, that the conditions of colour, smell and turbidity of the river below Teddington are so vile as compared with the Upper Thames, especially as to the above sources of filth must be added the tidal current, which is so rapid that it keeps the mud continually in suspension, washing it up at one time, depositing it at

another, but never permanently leaving it except in the places unscoured by the upland water, such as docks, backwaters and places out of the main current. It has been acknowledged by all writers that if the upland water should be stopped the Thames would become a stagnant oscillating ditch, because all filth discharged into it would remain in it permanently.

The docks trap a very large proportion of this mud, and it costs at least £60,000 per annum to clean it out. The mud enters with the locking water and with that pumped to make up the basins.

***Effect of Dockisation
on the River.***

It is proposed to construct across the river at Gravesend a dam or barrage similar to that across the Nile, containing numerous adjustable sluices, and in addition a series of very large locks, the dam to hold up the river to about Trinity high-water level (see section, [Fig. 4](#)).

The immediate effects will be these:—

(a) The tides, Neaps and Springs, will be stopped at the dam.

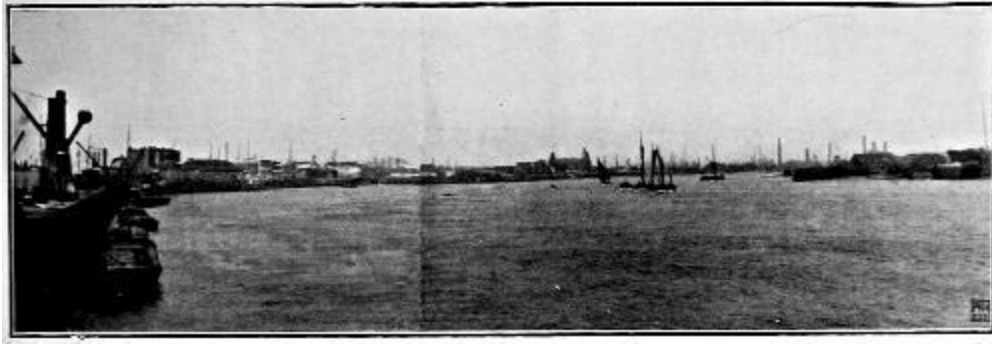
(b) The river will be converted into a long lake having numerous affluents, the principal of which will be its natural flow over Teddington Weir.

(c) It will have a slow downward current, never reversed, so that all that enters it will pass downwards to the dam.

(d) Its level (normally at Trinity high water) can be regulated to any level above low water by the sluices.

(e) Within from 25 to 45 days of the closing of the dam the upland water will have pushed over the dam all the oscillating foul water of the tidal river, and thenceforward the water of the lake will be the same as that of the upper river, and any soilage in it must enter it by sewage or land drainage.

(f) There will thus be obtained by one work a navigable depth of water varying from 65 ft. at Gravesend to 32 ft. at London Bridge, without dredging or any interference with the river bottom or banks.



THE RIVER THAMES BELOW BLACKWALL,
As it will appear when dockised.

But the consequent effects upon the business and usage of the river will be tremendous:—

(g) Ships drawing 30 ft. can then proceed to London Bridge at any hour of the day or night, without waiting for tides.

(h) Ships of all tonnages and draughts can traverse the river, anchor anywhere, lay alongside any wharf or quay, always remain at one level for loading or unloading (an immense boon to shipowners and wharf wharfingers) and need not lie out in the river or obstruct the free navigation.

(i) Dock entrances can be left open, thus saving the cost and time lost in working them. (The London and India Docks Co. estimates the cost of working their entrances at £50,000 per annum.)

(j) There will be no mud entering the docks and backwaters, the water in which will freely circulate with the clean river water.

(k) Exceptional tides, being stopped at the dam, will not overflow the river banks as now sometimes happens.

(l) Reduced cost of towage, barging, repairing river banks, camp-shedding, quays, dredging, management, control and policing of the river.

(*m*) Greatly increased safety of navigation: no grounding, swinging with the tides, collisions due to tidal drift. The tides are responsible for most of these accidents and for many lives lost—casualties which would not occur in a lake.

In addition to these there is a most valuable asset created in the advantage the new conditions open up for—

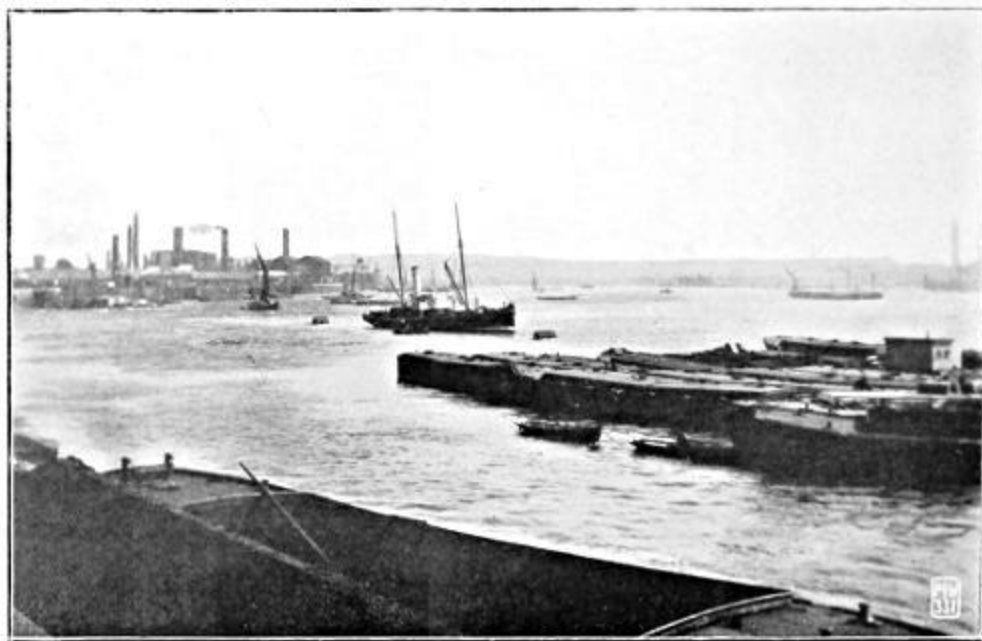
(*n*) Pleasure traffic, boating and sailing, fishing and the provision of efficient steamboat services, with fixed piers. London will be provided free with a lake of fresh water 45 miles long and from a quarter to half-a-mile wide. It is certain that this will give rise to extensive pleasure boating of all kinds, which will have ample room owing to the removal of all vessels from mid-stream anchorages to the shores.

The illustrations show the present crowded condition of some of the reaches of the river and the clearance that will be effected by a barrage.

Water Supply of London.

Perhaps the most important advantage created by the barrage will be the permanent supply of water for the increasing demands of the London area.

By the Act of 1903 has been created a Water Board which is empowered to purchase the water companies' properties and to administer them in the public interest. These companies claim £47,000,000 for their properties. The ratepayers pay them £3,000,000 annually for their water, and the companies pay £30,000 annually for the greater part of the water which they draw from the Thames.



BLACKWALL REACH.

The figures are as follows:—

	Gallons per day.
From the River Lea	52,500,000
” wells in the Lea Valley	40,000,000
” wells in the Kent Co.'s district	27,500,000
” the River Thames	185,000,000
Total	<u>305,000,000</u>

So that two-thirds of London's water supply comes from the Thames; and as the other sources named above cannot be expanded for future requirements, it is evident that for the increasing demands of London either the Thames or some more distant source must be looked to.

The Royal Commission on the water supply of London estimated that in 1941 these requirements will reach 423 million gallons per day, so that at that date 303 million gallons must be obtained from the Thames or elsewhere.

Now if the Thames is dockised, and the tides kept out of the river, it is evident that much less upland water than is now considered necessary will suffice to keep the river lake fresh and clean, because all sewage and effluents entering the river will be carried directly down to Gravesend; there will be no muddy foreshores and no stirring up of the river mud by the tidal scour.

The river will be, in fact, in exactly the same circumstances as most large lakes—that is, a large body of fresh water, having a main inlet of fresh water at one end, many small inlets along its banks, and one main outlet at its lower end at Gravesend. Such lakes abound all over the world: they are the purest of all waters and never become stagnant.

It is proposed, therefore, that the Thames lake should be regarded as a storage reservoir, so far as water supply is concerned. It will contain sufficient for 320 days' supply, even at the estimated requirements of 1941; for to whatever extent its waters may become contaminated at and below London, these pollutions cannot work back up the river towards Teddington. It follows, therefore, that between Teddington and London water may safely be drawn off for town supplies, or the supply may be taken as now from above Teddington.

An inspection of the table of flow over Teddington Weir on page 3 will show that in the winter and spring enormous quantities of water, above the quantity considered necessary for scouring the river, flow down and are lost.

A minimum flow of 200 million gallons is fixed by law as the amount needed in summer to keep some sort of cleanliness in the lower river; but in January ten times this amount flows away. It is only for a short time in the

months of August or September that the natural flow over Teddington Weir—including the water drawn by the water companies—is a little below 423 million gallons daily, and in those months the surplus might be taken from below the weir without affecting the river materially.

If this be objected to, however, there is another remedy available. The Upper Thames may be used as an aqueduct to convey a larger supply, to be derived from neighbouring watersheds or from wells, the water so obtained to be regulated to meet the requirements, enabling a sufficient amount to be run over the weir to keep the lower river in motion at its upper end. Further down, the small but numerous affluents and springs will keep the river in motion, as they are not affected by the Teddington flow, but give a continuous supply to the river. Mr. Topley, the eminent geologist, in his evidence before the London Water Commission, 1892, stated that there are outside the Thames basin large areas from which water could be obtained, such as East Kent, West Suffolk, Norfolk, Hampshire and Wilts.

It is evident that in this way an enormous prospective outlay for a supplementary water supply for London in the near future may be obviated, and that without adding to the existing plant of the water companies the new Water Board may inherit free of cost a future source of supply which will make their purchase of the London Water Companies' stocks a good investment and a cheap one for the ratepayers.

Rail and Road Communication at Gravesend.

The possibilities of this scheme are not exhausted, as there remains to be mentioned the opening of railway communication across the river by a tunnel under the dam and of road communication by a roadway over the dam. These are clearly shown in the accompanying [Figs. 4, 5 and 6](#).

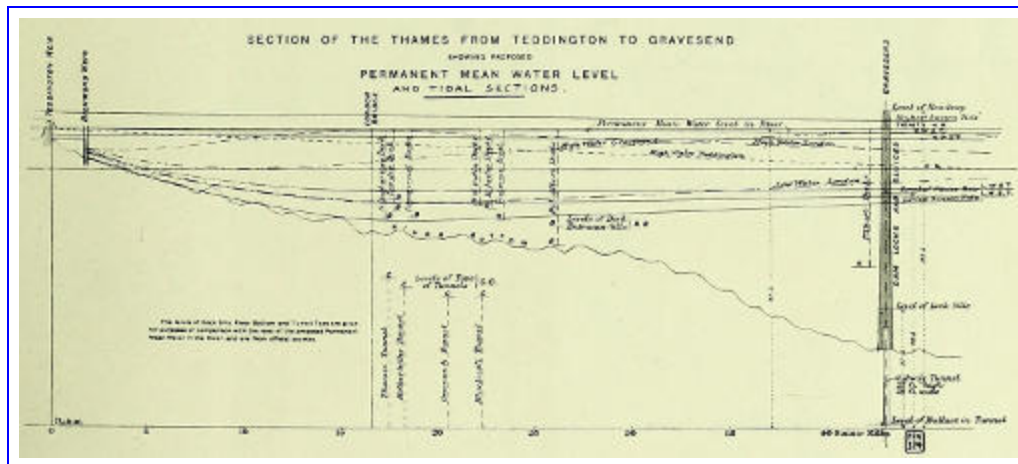
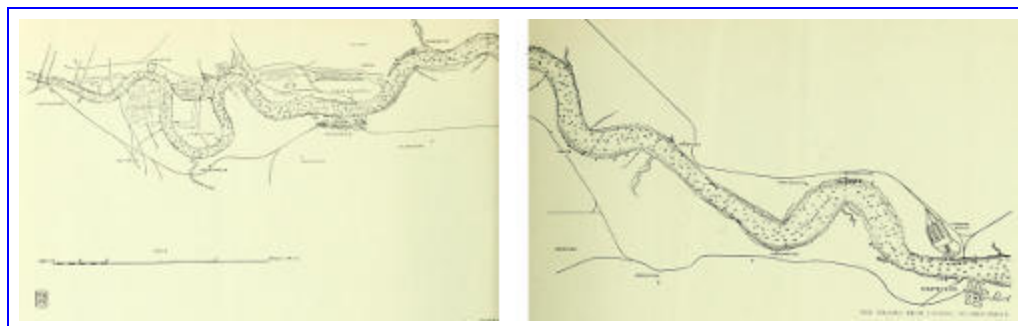


Fig. 4.

SECTION OF THE THAMES FROM TEDDINGTON TO GRAVESEND SHOWING PROPOSED PERMANENT MEAN WATER LEVEL AND TIDAL SECTIONS.



THE THAMES FROM LONDON TO GRAVESEND.



THE THAMES ESTUARY.

The tunnel will be constructed in the foundation of the dam, and the road formed on the top of the dam, and provided with opening bridges across the locks.

A glance at a railway map will at once show the strategic value of the railway route thus opened up between the Midlands and the North, and Dover and the South Coast, avoiding the congested London lines; also for national and military direct traffic between the Government arsenals and the Colchester and northern routes and depots. All the northern lines will thus have access by the Tilbury line to the continental routes.

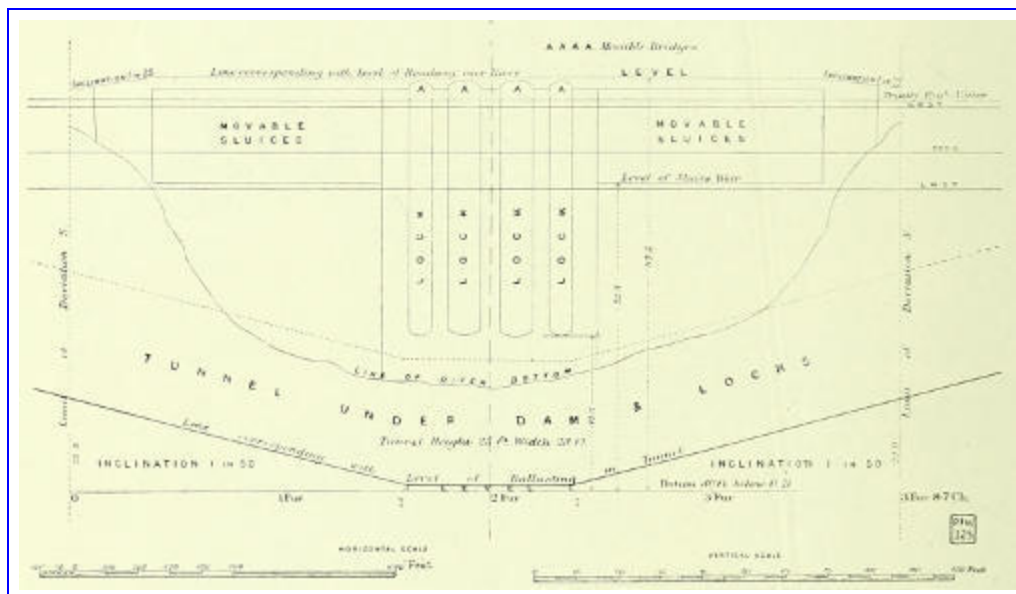


Fig. 5.

***National and Military
Aspect of the Scheme.***

The Port of London above the barrage will be the finest and safest harbour we possess for the fleet, having an immense deep-water protected area. The barrage can be fortified, and will constitute the most effective prevention against any foreign invasion by way of the Thames estuary. The tunnel and roadway will be of great service in this connection also.

***The Depletion of the
Thames Basin.***

This, which has been increasing for many years, is becoming a serious matter, and has attracted much comment. One of the advantages that will be obtained from the barrage will be the raising of the underground water-levels in the chalk and other strata of the Thames basin. In this way a permanent improvement in the water supply by wells throughout this large area will result.

Minor Advantages. Among these may be mentioned:—No further scouring of bridge or other foundations. No backing up of the foul waters of the small tributaries, such as the Lea, Barking Creek and others. Improved living conditions and reduction of disease, especially in the neighbourhood of the river, resulting from the cessation of ebb and flow, of smells and exposure of mud banks. Increased value of properties bordering the river. Fixed piers for passenger steamers.

Works and Construction.

[Fig. 7](#) is a general plan showing the barrage in relation to Tilbury and Gravesend shores.

[Fig. 5](#) is a cross section of the river showing the vertical dimensions and contours.

[Fig. 6](#) shows a section and details of construction.

Generally it is proposed to form the barrage of mass concrete, faced with granite on all exposed faces. The tunnel will be formed in the solid monolith as the work proceeds, and afterwards connected north and south with the existing railways. The foundation is in the chalk. The method of construction will be by cofferdam, to enclose an area sufficient for the walls and locks, which, when completed, can be opened for the up and down traffic of the river while the construction of the weirs and sluices is proceeded with. The sluices will be left open for the free passage of the tides until the closing of the barrage, which will take place at high water of a Spring tide.

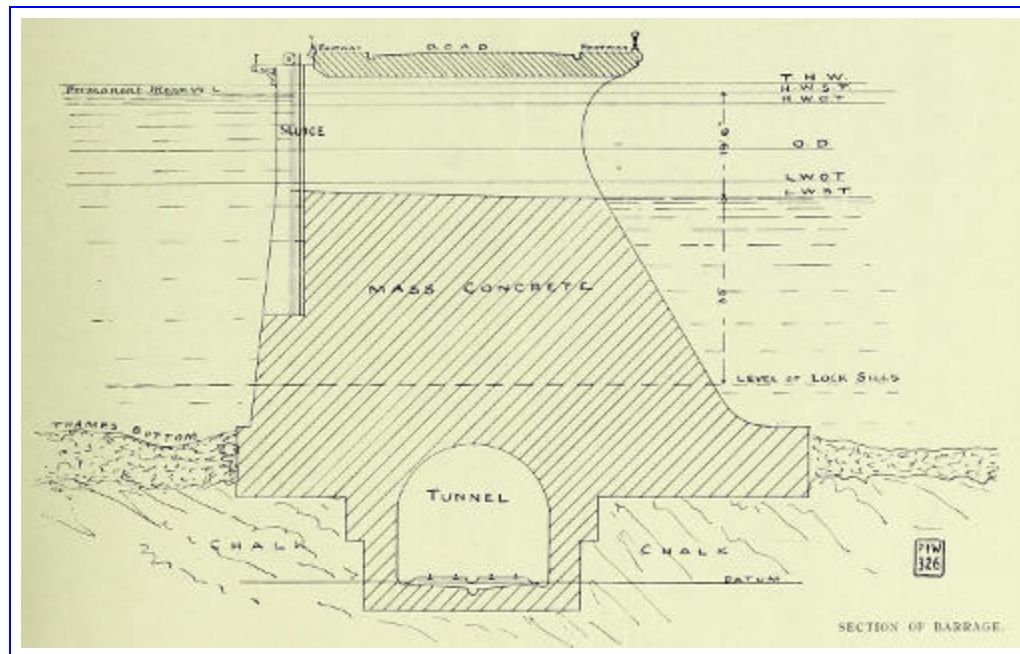


Fig. 6.

SECTION OF BARRAGE.

The locks will be worked electrically from a power-house built upon the central pier of the locks; the power to be obtained from dynamos operated

by the fall of part of the water flowing over the dam. A pilot tower will be fixed from which the river traffic will be signalled and regulated, and the locks, movable bridges, etc., controlled.

The locks as shown are four in number, each provided with internal gates in addition to the outer ones, in order that these locks may be worked in long or short lengths to suit the traffic. The lengths provided in this way will be 300 ft. 500 ft., 700 ft. and 1000 ft., and the widths 80 ft. and 100 ft. It is not likely that these dimensions will ever be exceeded by steamships.

The number of vessels passing up and down the river per day averages 220, but few of these exceed 300 ft. in length. It will be easy to lock this number up and down, or three times the number with this series of locks, one important advantage to the shipping being that, instead of waiting tides at Gravesend, each vessel as she arrives, at any hour, can be locked in a few minutes, up or down, without waiting.

Special provision will be made for rapidly and safely passing into and out of the locks with the use of power capstans and gear. The sluices will be of steel, sliding in roller guides, balanced and operated each by its own motor.

At or near low water a large volume of water will be sluiced into the lower river to scour the approach to the locks as often as found necessary.

A system of signalling from the Upper Thames to the barrage will be employed to notify any heavy rainfall or freshet coming down the river, so that by lowering the sluices water may be rapidly discharged to maintain the required level in the river, and at certain fixed dates it may be desirable to let down the water-level for a fixed time to allow of the repairing of dock entrances, walls, and other river-side works.

Financial.

The estimated cost of the barrage complete is £3,658,000, including compensations and other contingencies. A toll of $\frac{3}{4}$ d. per ton on the shipping passing up and down will pay the interest on this sum. This $\frac{3}{4}$ d. per ton additional toll will, it is estimated, be many times compensated for by reductions in the river and dock dues and other expenses, as below:

SAVINGS EFFECTED BY DOCKISATION.	Per Annum. £
Dredging in the river	200,000
Repairing banks, campsheds and groynes	10,100
Mudding in all docks	50,000
Cost of operating dock entrances and pumping	70,000
Saving in time of vessels ascending and descending the river	225,000
Saving in towage	20,000
" barging	185,000
" warping, buoying, lying off, etc.	20,000
" management of river	70,000
Total annual saving	£850,100

This is equal to a reduction of 6·8d. per ton on the tonnage of shipping (30,000,000) entering and leaving the Port, or equal to $7\frac{1}{2}$ times the interest on the cost of the barrage.

To the credit of the barrage must also be set the removal from the prospective future of enormous outlays contemplated for:—

	£
Purchasing docks, estimated at	30,000,000
Improving ditto and dredging river	7,000,000
Cost of a water supply from Wales or other source	24,000,000
Total	£61,000,000

***The Port of London
Bill, 1903.***

This measure is the Government's attempt to put into law the recommendations of the Royal Commission on the Port of London, 1902, but with amendments. It is proposed to purchase the entire docks and warehouses, leaving the wharves to run on their own resources; to create a Port Trust to control the entire river and docks; to charge the loan for purchase, etc., upon the London County Council—*i.e.*, about £35,000,000: and to dredge the river to about 30 ft. at low water up to the principal dock entrances.

Dredging the River.

Apart from its cost and the grossly unfair policy of financing and running the docks against the wharfingers, it is evident that this scheme is based upon the possibility of dredging the river to the depth required. [Fig. 8](#) is an actual section of the river, showing the proposed dredged channel as compared with a dockised river.

It seems incomprehensible that any expert authorities should have advised the Government that the river can be effectually dredged. The fact is that it is quite impossible to dredge it to the required depth of about 15 ft. below the present bottom, because experience has shown that with such a river and scouring current the channel will fill up again nearly as fast as it is dredged, the material coming from the foreshores and the estuary. This will give rise to dangerous slipping in of river banks and walls. The estimates of the cost of this dredging (£2,500,000) are therefore entirely misleading.

The present bottom is formed and stands at the natural angle of repose for its present volume, width and currents, and any great interference with this contour such as is proposed—with slopes of 7 to 1—will not stand, the general slope of its bottom now being from 20 to 50 to 1. The Port Trust that undertakes this will find itself spending enormous sums annually in continuous dredging and repairing banks and in compensating owners; all, of course, added to the annual cost of maintenance and to the dues, or charged to the ratepayers.

Glasgow and the Clyde have been instanced as examples of what can be done by dredging. But the Clyde below Glasgow is not a river comparable with the Thames below Gravesend, but an estuary with a very moderate current and tidal range of from about 4 ft. to 10 ft., and the dredging has merely made and kept open a channel in this estuary. The Thames, on the other hand, is a narrow river with a strong scouring current and a range of tide of from 16 ft. to 21 ft. Further than this, Glasgow has spent seven millions in this work, and has to pay large sums to keep the channel open, dredging nearly a million cubic yards every year.

But there are other difficulties. When the river has been deepened as proposed, the tidal volume will be increased about one-third, and therefore its current strengthened and increased, probably two knots per hour. What is

worse, the tidal range will be increased proportionately, which means that the high tides will be higher—probably 3 ft. or more—and the low tides lower, by a similar amount, than now. Spring tides may be expected to run the river nearly dry at low water above London Bridge. Results—frequent inundations of waterside districts, more grounding at low water, and more dangerous navigation. Such results have always followed increased tidal volume.

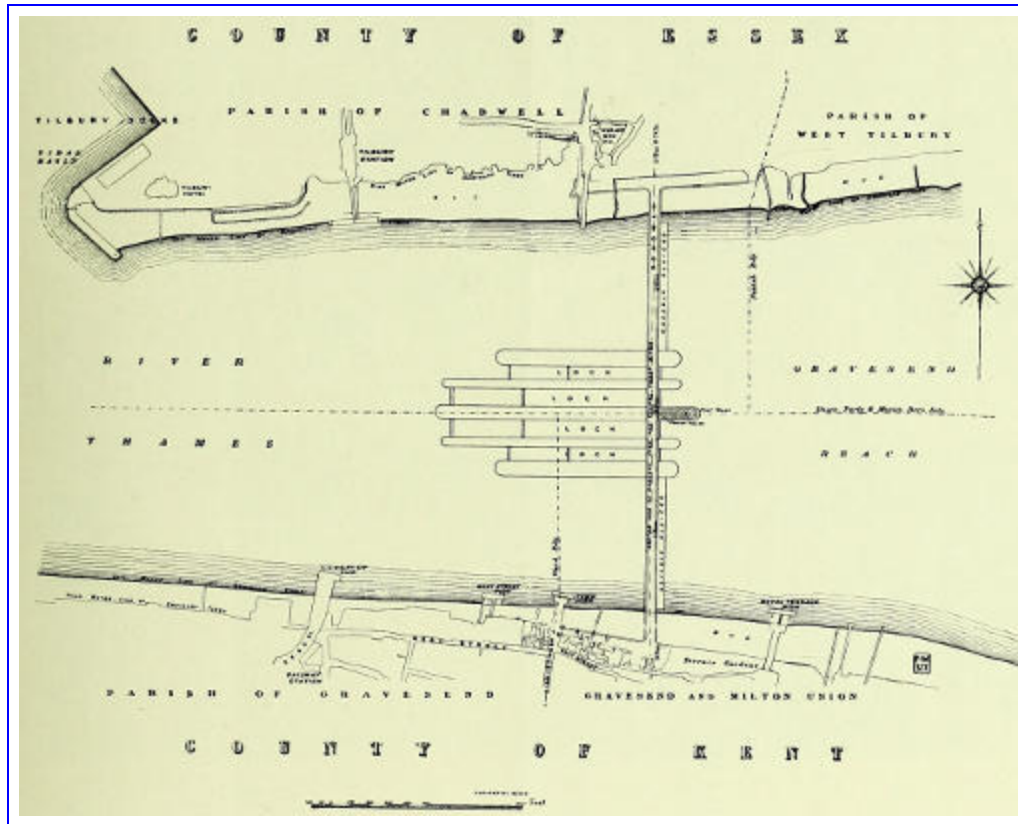


Fig. 7.

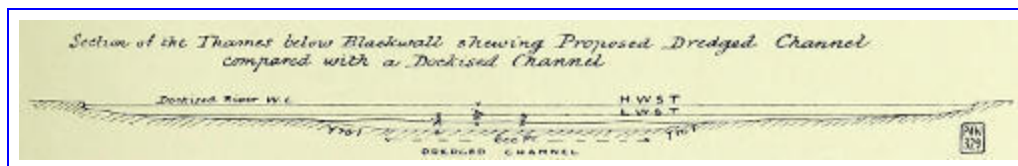


Fig. 8.

Section of the Thames below Blackwall shewing Proposed Dredged Channel compared with a Docked Channel.

But a dredged channel is necessarily a narrow one (see [Fig. 8](#)), and ships will have to negotiate the sharp bends in a narrow channel and against a

stronger tide, and also to swing at anchor, for which a wide area is necessary.

Objections against Dockisation.

Although this proposal has been mooted for some time past, scarcely any valid objection has been brought forward, but such as have been mentioned are mostly based on misconceptions.

One writer thought the river would become stagnant. As a matter of fact the sources of stagnation would be carried down the river by the fresh-water flow continuously, and there is no more reason to anticipate stagnation in the lower river than the upper river, where it has for ages been held up in the same way by numerous dams.

Another writer talks of the “cleansing power of the tides,” and it is a pity to see greater authorities, who ought to know better, speaking also in this way. It has been abundantly proved that the tides—as far as a clean river is concerned—are wholly detrimental. They back up twice daily the natural drainage of the river for five hours, and keep it in solution and circulation for forty-five days before removing it, the effect being exactly similar to backing up in a sewer.



THE POOL BELOW TOWER BRIDGE.

It has also been suggested that the sewage effluents discharged into the river at Crossness and Barking may cause the river below to become foul. Here again is misconception. The effluents—after precipitation of the solids, which is chemically effected, and the carrying out to sea of the

resulting sludge to the amount of two million tons annually—contain very little impurity (only seven grains per gallon), and it has been proved by Dr. Dupré that 9/10ths of this becomes oxidised and absorbed in the large volume of water between the discharge and Gravesend. It is well known that in the case of “sewage effluents poured into a sufficiently large volume of otherwise comparatively pure water, the dissolved organic matter contained in it disappears with remarkable rapidity” (Sir Alex. Binnie).

Another critic suggests that the lower river will soon silt up under the new conditions. Most persons—seeing the filthy state of the water—naturally think there must be a large deposit from it. But it has been shown that this suspended matter is the result of tidal currents keeping the mud stirred up everlastingly. An examination of the affluents of the Thames shows that they contain very little suspended matter, and therefore when the locked Thames has deposited its charge of suspended matter any future soilage must come from its affluents—that is, from the upland waters and the sewage effluents, which latter will only affect it below the point of their discharge.

A calculation from official data of the quantities actually now passing into the Thames, from all sources, gives less than 1/10th of an inch annually over the river bottom; so that in ten years the deposit will not exceed 1 in., even without any improvement in the prevention of pollution. It has been estimated by Dibdin that the sewage outfalls could be removed to Gravesend, below the barrage, for the sum of £4,000,000.

But the condition of these effluents is commonly much exaggerated. The total annual discharge of suspended matter at 7 grains per gallon (as given by Dibdin) amounts to 32,000 tons per annum, but much of this becomes chemically combined with the river water and some remains in suspension till it passes Gravesend, leaving only a small quantity to deposit in the river. A single dredger can remove 600 tons per hour; therefore a few hours' work will remove the whole quantity.

A more valid objection at first sight is that ships and barges will lose the motive power of the tides up and down. This would appear, however, to be a very beneficial loss, because at the same time they will avoid the tide-waiting and waste of time which add considerably to the cost of transit. But against this loss must be set the fact that most ships now have steam power

and can make their own destination, while tugs will be able to handle much larger fleets of barges than is now possible in the tide-way, and at all hours of the day. Sailing vessels will be able to sail up and down, which they can only do now with the aid of the tide.

Another suggestion is that when the barrage has closed the river the tides below it may accumulate to a higher level and overflow the low-lying lands below Gravesend. This is, however, a mistake, the fact being that with a reduced tidal volume and momentum in the estuary the tidal range will be reduced, there being no river to fill up, the high tides will be lower and the low tides higher than formerly.

Finally, a word or two as to the vague idea that seems to be in the minds of most people accustomed to tidal rivers—that in some mysterious way the tides by their continual movements are beneficial, keeping the air in motion, etc. All this is pure imagination and arises probably from living on the banks of a tidal river, for most rivers are non-tidal. There happen to be round our coasts some phenomenal ranges of tide; hence the resort to docks, which are almost unknown in other countries. The ranges of their tides being small, docks are not needed, and scarcely any tides occur in their rivers, which, however, are far cleaner than the Thames.

There are of course some low-lying lands bordering the river the drainage from which will have to be pumped into the river. This is, in fact, partially done now, but the matter is a small one.

Prof. Flinders Petrie, in a letter to the *Times*, is strongly in favour of this proposal, and looks to it to relieve the squalor of the East End, with its crowded and unhealthy living, by extending the manufacturing districts down the river banks, providing a belt of factories along each bank and a belt of garden villages behind them, with fast lines of railway to Town between.

To carry out the proposals of this article, a committee has been formed to bring the subject before the notice of Parliament and of the public, and it is suggested that a Board of Harbour Commissioners should be formed, somewhat on the lines of the Port of London Bill of last Session. The new Board would be constituted under the usual Commissioners' Acts to control the entire Lower Thames, taking over the powers of the existing authorities, but without any interference with the docks, the warehouses or the wharves,

the business of which, if the river is rendered properly navigable, could be carried on without making any demands upon the rates of London.

A new era of prosperity would then open up for the trade of London, and its Port would become the finest in the world, with the largest business attached to it.

The committee will include many influential gentlemen connected with and interested in the improvement of the Port of London. The scheme originated with Mr. Jas. Casey, M.I.N.A., and the author is responsible for the engineering details, as also for the information set forth in the foregoing article.



*** END OF THE PROJECT GUTENBERG EBOOK THE GREAT
THAMES BARRAGE ***

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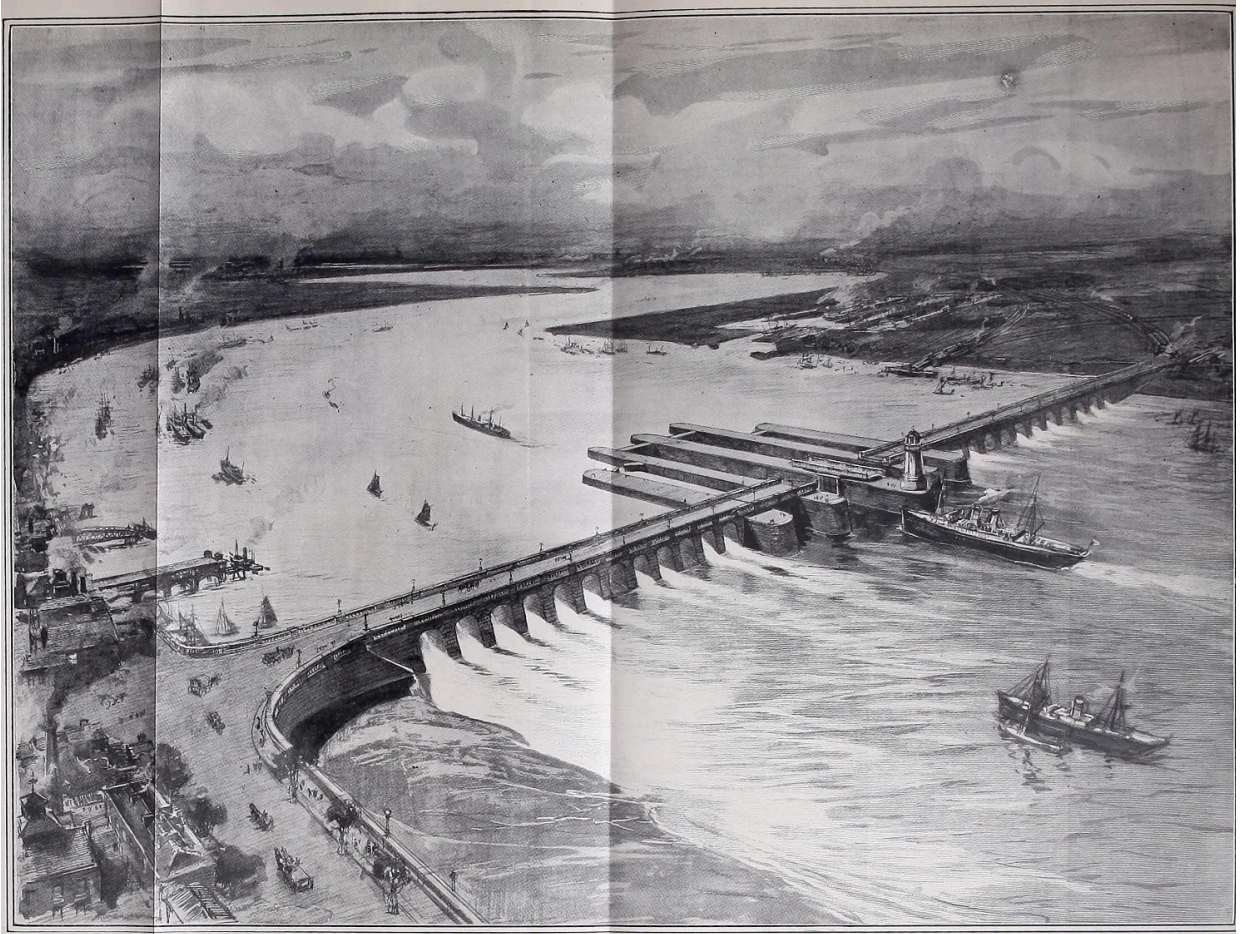
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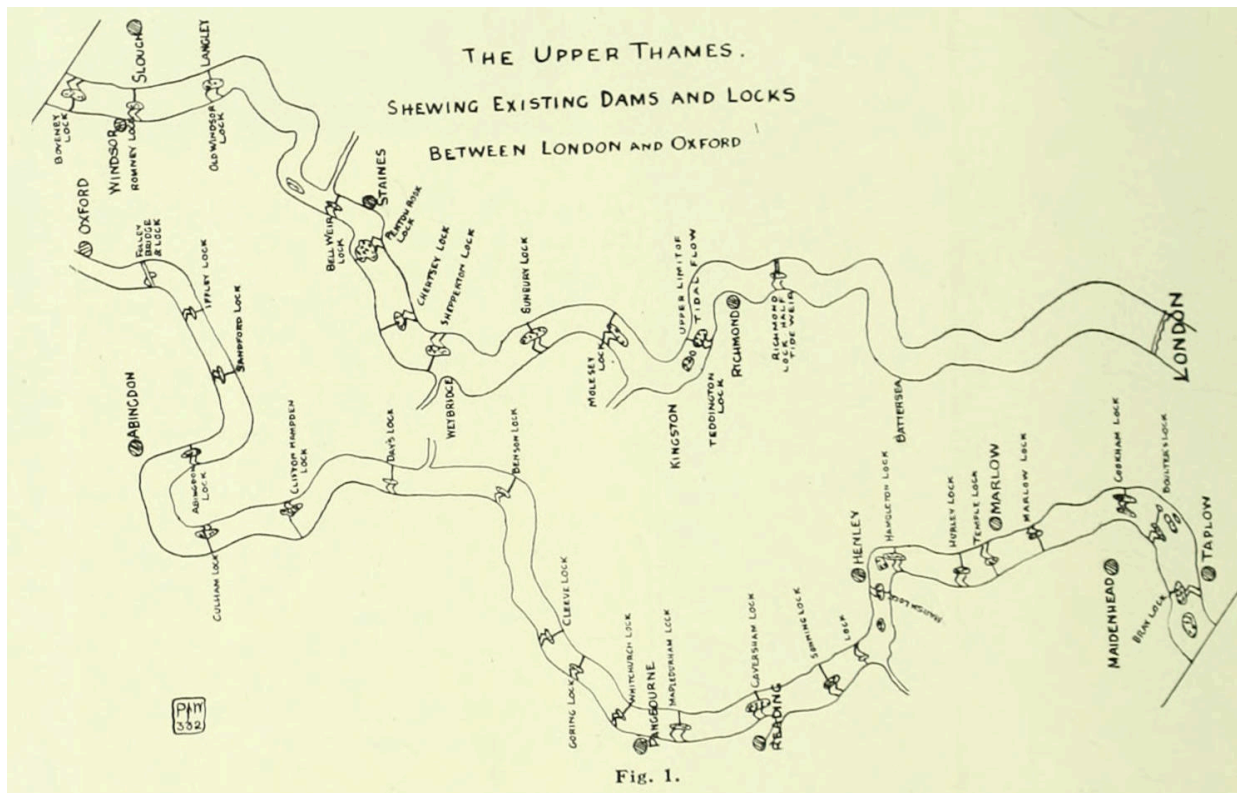
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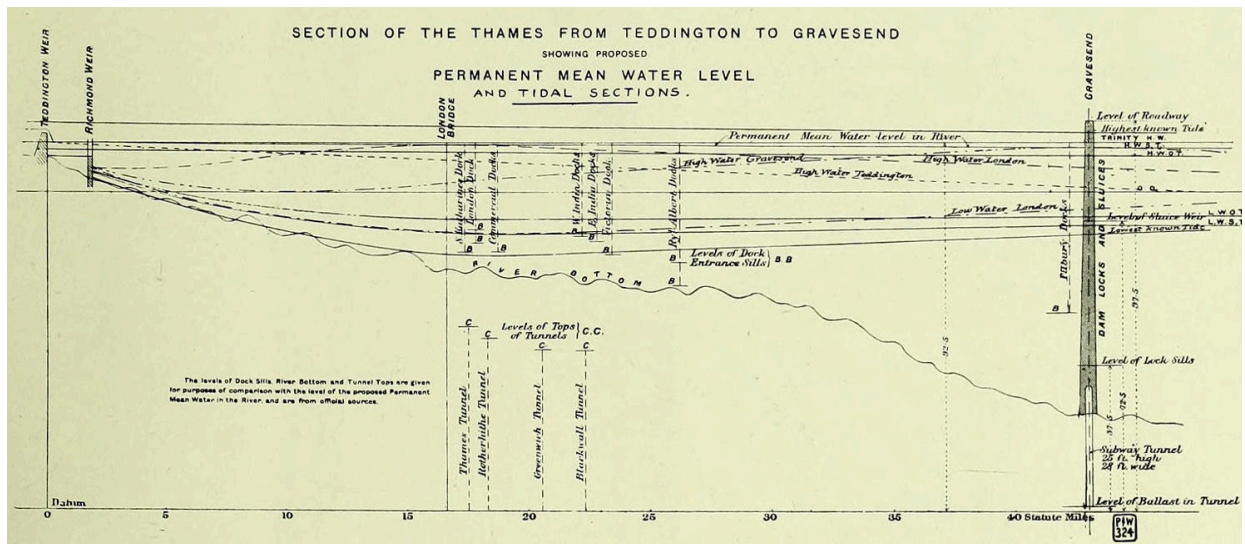
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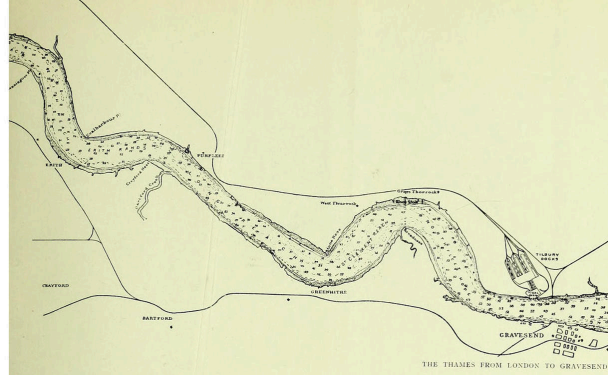
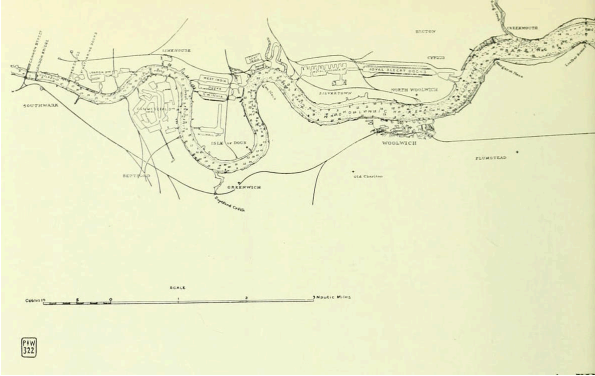
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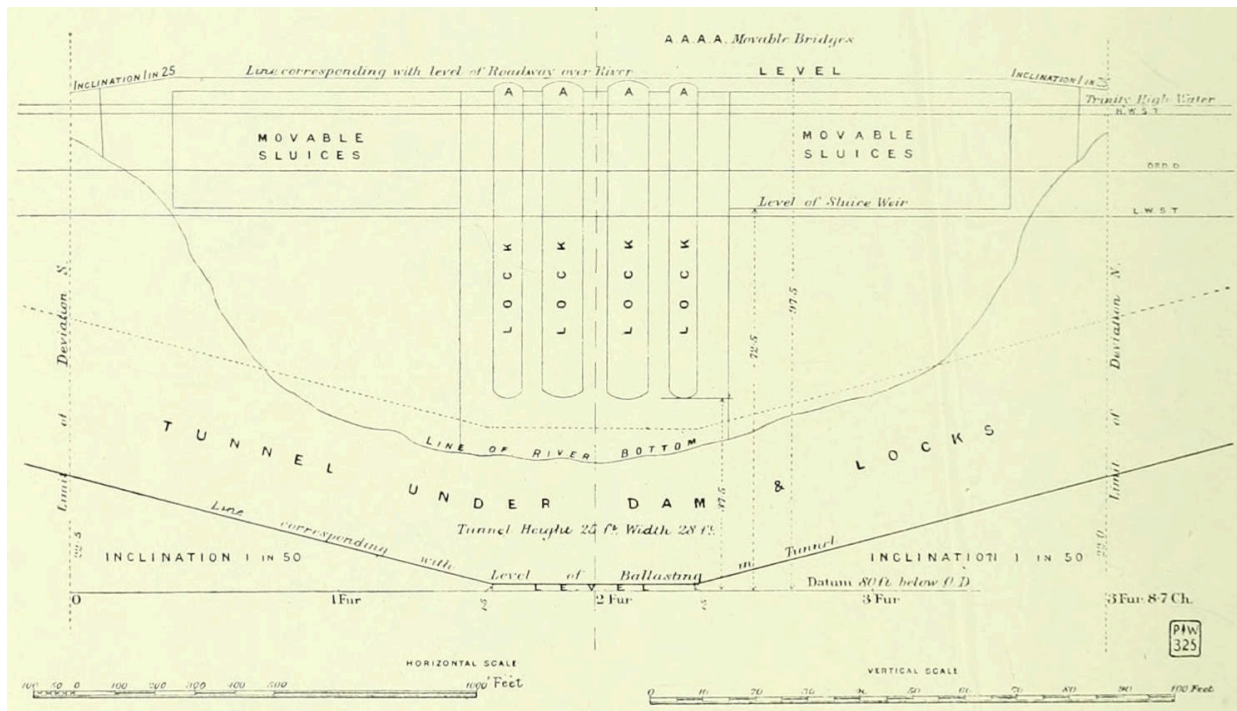
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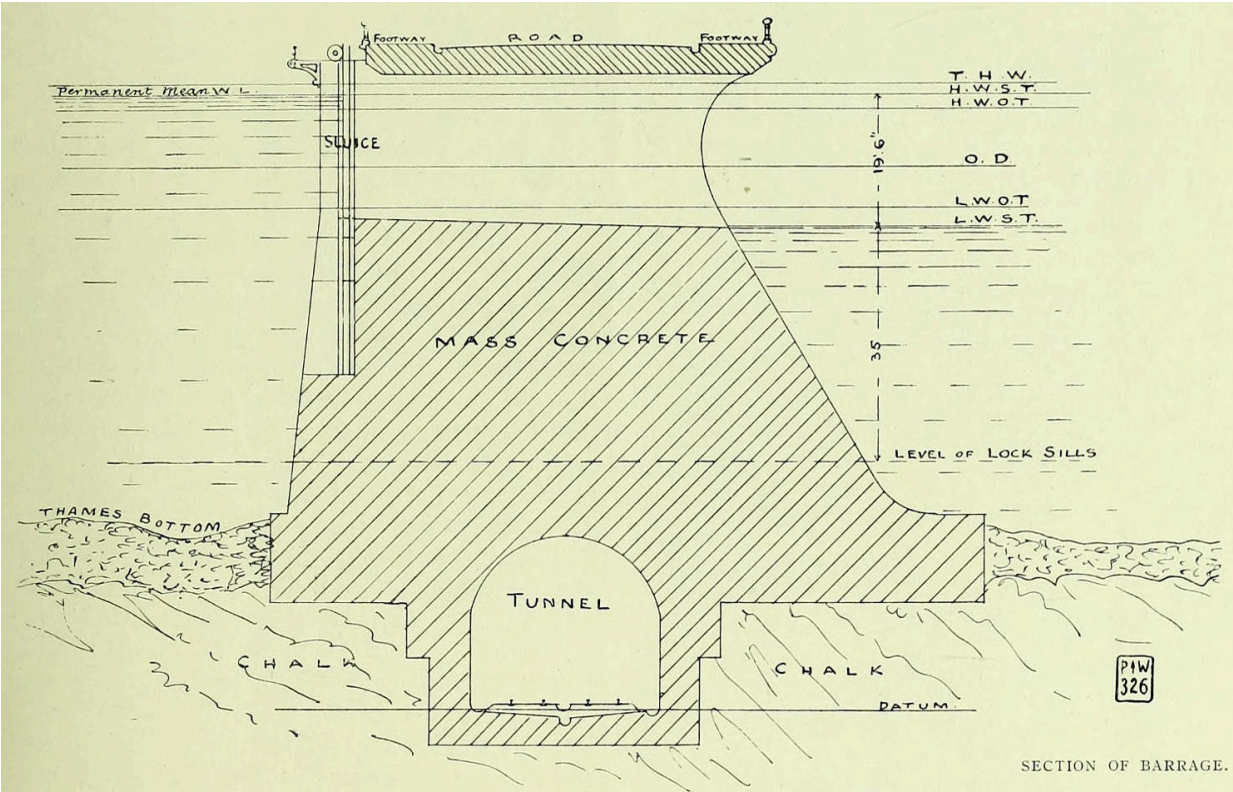
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